SALEDUCATION

SALENGINEERINGANDTECHNICALINSTITUTE

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Student's Weekly Reportof Internship

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Work done in last Week with description :

Week 10 (Date :29th March 2024 to 4th April 2024) :

1) Malware Analysis

Plans for next week:

1) Remnux and Flare vm

Task

- **Malware Analysis**

- Malware analysis is the process of examining malicious software (malware) to understand its behavior, functionality, and impact on systems and networks. Malware analysis is a critical component of cybersecurity and is performed by security researchers, malware analysts, incident responders, and forensic investigators to identify, analyze, and mitigate cyber threats. Here's a detailed explanation of malware analysis:

- Types of Malware Analysis:a.

- Static Analysis: Static analysis involves examining the malware without executing it. It typically includes examining the malware's code, file structure, metadata, and other attributes to identify indicators of compromise (IOCs), such as file hashes, file names, embedded strings, and cryptographic signatures. Static analysis techniques include examining file headers, disassembling code, decompiling binaries, and examining network traffic.

- Dynamic Analysis: Dynamic analysis involves executing the malware in a controlled environment (sandbox) and observing its behavior at runtime. It includes monitoring system calls, file system activities, network communications, process interactions, and registry modifications to understand the malware's functionality, capabilities, and impact on the system. Dynamic analysis helps identify malicious behaviors, such as file encryption, data exfiltration, privilege escalation, and command-and-control (C2) communications.

- Hybrid Analysis: Hybrid analysis combines static and dynamic analysis techniques to gain a comprehensive understanding of malware. It leverages both static and dynamic analysis tools and methodologies to identify malware characteristics, behaviors, and attack vectors effectively.

- Malware Analysis Techniques:

- Code Analysis: Code analysis involves examining the malware's code to understand its logic, functionality, and attack techniques. It includes disassembling or decompiling the malware binaries to analyze assembly instructions, function calls, control flow, and data structures. Code analysis helps identify malicious routines, evasion techniques, and vulnerabilities exploited by the malware.

- Behavioral Analysis: Behavioral analysis involves observing the malware's behavior in a controlled environment to understand its actions and impact on the system. It includes monitoring system activities, network traffic, file system modifications, registry changes, and process interactions to identify malicious behaviors, persistence mechanisms, and communication patterns.

- Signature Analysis: Signature analysis involves comparing the malware's characteristics, such as file hashes, file names, and code patterns, against known malware signatures in antivirus and threat intelligence databases. It helps identify known malware variants and associate them with existing threat categories and families.

- Protocol Analysis: Protocol analysis involves analyzing network traffic generated by the malware to identify communication protocols, command-and-control (C2) channels, and data exfiltration techniques. It includes capturing and inspecting network packets, decoding network protocols, and reconstructing communication flows to understand the malware's network behavior.

- Malware Analysis Tools:

- Static Analysis Tools: Static analysis tools include disassemblers (IDA Pro, Ghidra), decompilers (RetDec, jadx), binary analysis frameworks (Radare2, Binary Ninja), and static code analyzers (YARA, PEiD). These tools help analyze the malware's code, structure, and characteristics without executing it.

- Dynamic Analysis Tools: Dynamic analysis tools include sandboxing platforms (Cuckoo Sandbox, Joe Sandbox), behavior-based analyzers (ProcMon, Wireshark), malware analysis sandboxes (Hybrid Analysis, VirusTotal), and virtualization software (VMware, VirtualBox). These tools allow researchers to execute malware in a controlled environment and observe its behavior at runtime.

- Memory Analysis Tools: Memory analysis tools include memory forensics frameworks (Volatility, Rekall), memory dump analyzers (WinDbg, Redline), and memory visualization tools (MemProcFS, Memoryze). These tools help analyze malware artifacts in system memory, such as process memory, kernel data structures, and network buffers.

- Malware Analysis Process:
- Preparation: Prepare the analysis environment, including setting up isolated systems or virtual machines, installing analysis tools and utilities, and establishing network monitoring and logging capabilities.

- Collection: Collect malware samples from various sources, such as email attachments, web downloads, or infected systems. Gather information about the malware, including file hashes, file names, file sizes, and timestamps.
- Analysis: Analyze the malware using static and dynamic analysis techniques to understand its behavior, functionality, and impact on systems and networks. Use malware analysis tools and methodologies to identify indicators of compromise (IOCs), malicious behaviors, and attack techniques.
- Reporting: Document the findings of the malware analysis, including detailed descriptions of the malware's characteristics, behaviors, and capabilities. Generate comprehensive reports with analysis results, artifacts, and recommendations for remediation and mitigation.

Total Working Hrs: ~~**25 Hours**~~ **Signature of Student:**

The above entries are correct and the grading of work done by Trainee is

Excellent	Very Good	Good	Fair	Below Average	Poor

Signature of External Guide with Company Seal: **Signature of Internal Guide**

Date: **Date:**